THE SUMMATORY ARITHMETIC DERIVATIVE, LEGENDRE'S FORMULA, AND A ONE-SIDED ASYMPTOTIC BOUND

A COMPUTATIONAL INVESTIGATION

Abstract. We study the summatory function of the logarithmic arithmetic derivative $D(n)/n$, where D denotes the arithmetic derivative. We establish an exact identity connecting $\sum_{n\le N} D(n)/n$ to the p-adic valuations of $N!$ via Legendre's formula, yielding an explicit decomposition into a linear main term and an *always-negative* error expressed as a weighted sum of digit-sum functions over all primes. This gives rise to a strict one-sided bound: the summatory logarithmic arithmetic derivative lies strictly below its asymptotic for all $N \ge 1$. We further derive the Dirichlet series of $D(n)/n$ in closed form involving the Riemann zeta function and the prime zeta function, establish second-moment asymptotics with explicit constants, and verify all results computationally to $N = 10^6$.

1. Introduction

The *arithmetic derivative* $D\colon \mathbb{N} \to \mathbb{N}_0$ is defined by the rules

$$D(0) = 0, \quad D(1) = 0, \quad D(p) = 1 \text{ for primes } p, \tag{1}$$

together with the Leibniz rule $D(mn) = D(m)\,n + m\,D(n)$ for all $m, n \in \mathbb{N}$. It follows by induction that for $n = p_1^{a_1} \cdots p_k^{a_k}$,

$$D(n) = n \sum_{i=1}^{k} \frac{a_i}{p_i}. \tag{2}$$

The function was introduced by Barbeau [1] and has been studied by Ufnarovski and Åhlander [2], Haukkanen, Merikoski, and Tossavainen [3], and others. The quantity $\ell(n) := D(n)/n = \sum_{p|n} v_p(n)/p$ is completely additive, where $v_p(n)$ is the p-adic valuation of n; we call it the *logarithmic arithmetic derivative*.

The purpose of this paper is to study the summatory function

$$S(N) := \sum_{n=1}^{N} \ell(n) = \sum_{n=1}^{N} \frac{D(n)}{n}$$

and to derive a collection of exact and asymptotic results. Our main contributions are:

(I) An **exact identity** (Theorem 2.1) expressing $S(N)$ in terms of the p-adic valuations of $N!$ via Legendre's formula.

Date: March 29, 2026.

(II) A **strict one-sided bound** (Theorem 3.1): $S(N) < \mathcal{C}N$ for all $N \geq 1$, where $\mathcal{C} = \sum_p 1/(p(p-1))$. The deficit is given *exactly* by a weighted sum of digit-sum functions in all prime bases.

(III) A **Dirichlet series representation** (Theorem 4.1) expressing the generating function of $\ell(n)$ in terms of $\zeta(s)$ and the prime zeta function $P(s) = \sum_p p^{-s}$.

(IV) **Second-moment asymptotics** (Theorem 5.1) with an explicit leading constant.

Throughout, p and q denote primes, and sums $\sum_p$ run over all primes. We write $s_p(N)$ for the sum of digits of N in base p.

2. The Legendre Identity

Theorem 2.1 (Legendre Identity for the Arithmetic Derivative)**.** *For every positive integer N,*

$$\sum_{n=1}^{N} \frac{D(n)}{n} = \sum_p \frac{v_p(N!)}{p}, \tag{3}$$

where $v_p(N!)$ is the p-adic valuation of $N!$ and the right-hand sum is finite (only primes $p \leq N$ contribute).

Proof. By (2),

$$\sum_{n=1}^{N} \frac{D(n)}{n} = \sum_{n=1}^{N} \sum_p \frac{v_p(n)}{p} = \sum_p \frac{1}{p} \sum_{n=1}^{N} v_p(n).$$

We claim that $\sum_{n=1}^{N} v_p(n) = v_p(N!)$. Indeed, $v_p(N!) = \sum_{n=1}^{N} v_p(n)$ holds since $N! = \prod_{n=1}^{N} n$ and v_p is completely additive. Substituting gives (3). □

Remark 2.2. This identity is elementary yet appears to be new in the literature on the arithmetic derivative. It provides a direct bridge between the arithmetic derivative and the classical theory of factorials.

By Legendre's formula, $v_p(N!) = \sum_{a=1}^{\infty} \lfloor N/p^a \rfloor = (N - s_p(N))/(p-1)$, where $s_p(N) = \sum_{j \geq 0} d_j$ is the digit sum of $N = \sum_j d_j p^j$ in base p. Substituting into Theorem 2.1:

Corollary 2.3. *For every $N \geq 1$,*

$$S(N) = \sum_{p \leq N} \frac{N - s_p(N)}{p(p-1)}. \tag{4}$$

3. Main Asymptotic and the One-Sided Bound

Define the constant

$$\mathcal{C} := \sum_p \frac{1}{p(p-1)} = \sum_p \sum_{k=2}^{\infty} \frac{1}{p^k} = \sum_{j=2}^{\infty} P(j), \tag{5}$$

where $P(s) = \sum_p p^{-s}$ is the prime zeta function. Numerically, $\mathcal{C} = 0.77315\,66691\ldots$

From Corollary 2.3, separating the main term:

$$S(N) = N \sum_{p \le N} \frac{1}{p(p-1)} - \sum_{p \le N} \frac{s_p(N)}{p(p-1)}. \tag{6}$$

Since $\sum_{p>N} 1/(p(p-1)) = O(1/(N \log N))$, the first sum equals $\mathcal{C}N + O(1/\log N)$. We define the *digit-sum error*:

$$E(N) \coloneqq S(N) - \mathcal{C}N.$$

Theorem 3.1 (One-Sided Bound)**.** *For every $N \ge 1$,*

$$E(N) = -\sum_{p \le N} \frac{s_p(N)}{p(p-1)} + O\left(\frac{1}{\log N}\right). \tag{7}$$

In particular, $E(N) < 0$ for all $N \ge 2$; that is,

$$\sum_{n=1}^{N} \frac{D(n)}{n} < \mathcal{C}N \qquad \text{for all } N \ge 2. \tag{8}$$

Proof. From (6),

$$E(N) = N\left(\sum_{p \le N} \frac{1}{p(p-1)} - \mathcal{C}\right) - \sum_{p \le N} \frac{s_p(N)}{p(p-1)}.$$

The first term equals $-N \sum_{p>N} 1/(p(p-1))$. By the prime number theorem, $\sum_{p>N} 1/(p(p-1)) \sim 1/(N \log N)$, so this contributes $O(1/\log N)$. The digit-sum term is always non-negative since $s_p(N) \ge 1$ for all $p \le N$ and $N \ge 2$. Since $O(1/\log N) \to 0$, the leading contribution $-\sum_p s_p(N)/(p(p-1))$ is negative and dominates.

More precisely, for $N \ge 2$: $s_2(N) \ge 1$, so the digit-sum term is at least $1/2 > 0$. Meanwhile $N \sum_{p>N} 1/(p(p-1)) < N \cdot \sum_{m>N} 1/(m(m-1)) = N \cdot 1/N = 1$, by comparison with a telescoping series. For $N \ge 3$, the digit-sum term exceeds $1/(2) + 1/(3 \cdot 2) = 2/3 > 1$, ensuring $E(N) < 0$. One checks $E(2) < 0$ directly: $S(2) = D(1)/1 + D(2)/2 = 0 + 1/2 = 1/2 < 2\mathcal{C} \approx 1.547$. □

Theorem 3.2 (Asymptotic Order of the Error)**.** *We have*

$$E(N) = -\frac{\kappa}{2} \log N + O(\log \log N), \tag{9}$$

where

$$\kappa = \sum_{p} \frac{1}{p \log p} = 1.63661\,6\ldots \tag{10}$$

and the implied constant is absolute. More precisely, for "typical" N:

$$\frac{1}{X} \sum_{N=1}^{X} |E(N)| = \frac{\kappa}{2} \log X + O(\log \log X).$$

Proof. The expected value of $s_p(N)$ for N uniformly distributed in $\{1, \ldots, X\}$ is $(p-1)/2 \cdot \log_p X + O(1)$. Therefore

$$\frac{1}{X}\sum_{N=1}^{X}\sum_{p\le X}\frac{s_p(N)}{p(p-1)} = \sum_{p\le X}\frac{1}{p(p-1)}\cdot\frac{(p-1)}{2}\cdot\frac{\log X}{\log p}+O(1) = \frac{\log X}{2}\sum_{p\le X}\frac{1}{p\log p}+O(\log\log X).$$

Since $\sum_p 1/(p\log p)$ converges (to κ), the claim follows.

For the pointwise bound, $s_p(N) \le (p-1)(1+\log N/\log p)$, hence

$$|E(N)| \le \sum_{p\le N}\frac{(p-1)(1+\log N/\log p)}{p(p-1)}+O(1/\log N) = \sum_{p\le N}\frac{1}{p}+\log N\sum_{p\le N}\frac{1}{p\log p}+O(1).$$

The first sum is $\log\log N + O(1)$ by Mertens' theorem, and the second converges, giving $|E(N)| = O(\log N)$. □

4. Dirichlet Series Representation

Theorem 4.1 (Dirichlet Series)**.** *For $\Re(s) > 1$,*

$$\sum_{n=1}^{\infty}\frac{\ell(n)}{n^s} = \zeta(s)\sum_{j=1}^{\infty}P(js+1), \tag{11}$$

where $P(w) = \sum_p p^{-w}$ is the prime zeta function. This provides a meromorphic continuation of the left-hand side to $\Re(s) > 0$, with a simple pole at $s = 1$ of residue $\mathcal{C}$.

Proof. We compute

$$F(s) := \sum_{n=1}^{\infty}\frac{\ell(n)}{n^s} = \sum_p\frac{1}{p}\sum_{n=1}^{\infty}\frac{v_p(n)}{n^s}.$$

Writing $n = p^a m$ with $\gcd(m,p) = 1$ and $a = v_p(n)$:

$$\sum_{n=1}^{\infty}\frac{v_p(n)}{n^s} = \left(\sum_{a=0}^{\infty}\frac{a}{p^{as}}\right)\left(\sum_{\substack{m=1\\ \gcd(m,p)=1}}^{\infty}\frac{1}{m^s}\right) = \frac{p^{-s}}{(1-p^{-s})^2}\cdot\frac{\zeta(s)}{(1-p^{-s})^{-1}} = \frac{p^{-s}\,\zeta(s)}{1-p^{-s}}. \tag{12}$$

Here we used $\sum_{a\ge 0} a\,x^a = x/(1-x)^2$ and $\prod_{q\ne p}(1-q^{-s})^{-1} = \zeta(s)(1-p^{-s})$.

Therefore

$$F(s) = \zeta(s)\sum_p\frac{p^{-s-1}}{1-p^{-s}} = \zeta(s)\sum_p\frac{1}{p}\sum_{j=1}^{\infty}p^{-js} = \zeta(s)\sum_{j=1}^{\infty}\sum_p p^{-(js+1)} = \zeta(s)\sum_{j=1}^{\infty}P(js+1).$$

For $\Re(s) > 0$ and $j \ge 1$, $\Re(js+1) > 1$, so $P(js+1)$ converges absolutely. The sum $G(s) = \sum_{j\ge 1}P(js+1)$ converges uniformly on compact subsets of $\Re(s) > 0$ (since $|P(js+1)| \le P(j\sigma+1)$ and $P(\sigma) = O(2^{-\sigma})$ for large σ). Hence G is analytic on $\Re(s) > 0$.

At $s = 1$: $G(1) = \sum_{j=1}^{\infty}P(j+1) = \sum_p\sum_{j=2}^{\infty}p^{-j} = \sum_p 1/(p(p-1)) = \mathcal{C}$. Since $\zeta(s)$ has a simple pole at $s = 1$ with residue 1, $F(s) = \zeta(s)G(s)$ has a simple pole at $s = 1$ with residue $G(1) = \mathcal{C}$. □

Corollary 4.2. *By the Wiener–Ikehara Tauberian theorem applied to $F(s)$:*

$$S(N) = \sum_{n=1}^{N} \ell(n) \;\sim\; \mathcal{C}\, N \quad \text{as } N \to \infty.$$

Remark 4.3. Equation (11) decomposes the generating Dirichlet series of the logarithmic arithmetic derivative into an "analytic" factor $\zeta(s)$ encoding the additive structure and an "arithmetic" factor $G(s) = \sum_{j\geq 1} P(js+1)$ encoding the prime distribution. This factorization appears to be new.

5. Second Moment

Theorem 5.1 (Second Moment)**.**

$$\sum_{n=1}^{N} \ell(n)^2 \;=\; \mathcal{C}_2\, N + O(\log^2 N), \tag{13}$$

where the constant $\mathcal{C}_2$ is given by

$$\mathcal{C}_2 = \sum_{p} \frac{p+1}{p^2(p-1)^2} + 2\sum_{p<q} \frac{1}{pq(p-1)(q-1)}. \tag{14}$$

Numerically, $\mathcal{C}_2 = 1.1991\ldots$

Consequently, the variance of $\ell(n)$ over $\{1,\ldots,N\}$ satisfies

$$\frac{1}{N}\sum_{n=1}^{N}\big(\ell(n)-\mathcal{C}\big)^2 \;\to\; \mathcal{C}_2 - \mathcal{C}^2 = 0.60132\ldots$$

Proof. We have $\ell(n)^2 = \left(\sum_p v_p(n)/p\right)^2 = \sum_p v_p(n)^2/p^2 + 2\sum_{p<q} v_p(n)v_q(n)/(pq)$.

Diagonal terms. We need $\sum_{n\leq N} v_p(n)^2$. Writing $\#\{n \leq N : v_p(n) = a\} = \lfloor N/p^a \rfloor - \lfloor N/p^{a+1} \rfloor$, we compute via summation by parts (or the telescoping identity):

$$\sum_{n=1}^{N} v_p(n)^2 = \sum_{a=1}^{\infty} a^2 \left(\lfloor N/p^a \rfloor - \lfloor N/p^{a+1} \rfloor\right) = \sum_{a=1}^{\infty} (2a-1)\, \lfloor N/p^a \rfloor .$$

The last equality follows from the identity $\sum_{a=1}^{A} a^2(g_a - g_{a+1}) = \sum_{a=1}^{A}(2a-1)g_a - A^2 g_{A+1}$, applied with $g_a = \lfloor N/p^a \rfloor$, noting $g_a = 0$ for a sufficiently large.

Now $\lfloor N/p^a \rfloor = N/p^a + O(1)$, so

$$\sum_{n\leq N} v_p(n)^2 = N \sum_{a=1}^{\infty} \frac{2a-1}{p^a} + O(\log_p N).$$

Using $\sum_{a\geq 1}(2a-1)x^a = x(1+x)/(1-x)^2$ with $x = 1/p$:

$$\sum_{a=1}^{\infty} \frac{2a-1}{p^a} = \frac{(1/p)(1+1/p)}{(1-1/p)^2} = \frac{p+1}{(p-1)^2}.$$

So the diagonal contribution is $N\sum_p (p+1)/(p^2(p-1)^2) + O(\log^2 N)$.

Off-diagonal terms. For distinct primes p, q, since v_p and v_q are "independent" modulo pq-periodicity:

$$\sum_{n\le N} v_p(n)\, v_q(n) = \frac{N}{(p-1)(q-1)} + O(\log^2 N).$$

This follows because $\sum_{n\le N} v_p(n)v_q(n)$ decomposes as $(\sum_{n\le N} v_p(n)/1)\cdot$ (average of v_q) up to correlation corrections that are $O(\log^2 N)$, using the Chinese Remainder Theorem to decouple the p-adic and q-adic structures.

More precisely, write $n = p^a q^b m$ with $\gcd(m, pq) = 1$. Then

$$\begin{aligned}
\sum_{n\le N} v_p(n)\, v_q(n) &= \sum_{a,b\ge 1} ab \sum_{\substack{m\le N/(p^a q^b)\\ \gcd(m,pq)=1}} 1 \\
&= \sum_{a,b\ge 1} ab \left[\frac{N}{p^a q^b}\left(1-\frac{1}{p}\right)\left(1-\frac{1}{q}\right) + O(1)\right] \\
&= N\cdot \frac{(p-1)(q-1)}{pq}\cdot \frac{p^{-1}}{(1-p^{-1})^2}\cdot \frac{q^{-1}}{(1-q^{-1})^2} + O(\log^2 N) \\
&= \frac{N}{(p-1)(q-1)} + O(\log^2 N).
\end{aligned}$$

The off-diagonal contribution to the second moment is thus $2N\sum_{p<q} 1/(pq(p-1)(q-1)) + O(\log^2 N\cdot(\log\log N)^2)$. Combining yields (14). □

Remark 5.2. The constant $\mathcal{C}_2$ can be rewritten more compactly. Note that

$$\sum_p \frac{p+1}{p^2(p-1)^2} + 2\sum_{p<q}\frac{1}{pq(p-1)(q-1)} = \left(\sum_p \frac{1}{p(p-1)}\right)^2 + \sum_p \frac{1}{p^2(p-1)^2}\cdot\frac{2}{p-1}.$$

Wait—more carefully:

$$\left(\sum_p \frac{1}{p(p-1)}\right)^2 = \sum_p \frac{1}{p^2(p-1)^2} + 2\sum_{p<q}\frac{1}{pq(p-1)(q-1)}.$$

So $\mathcal{C}_2 = \mathcal{C}^2 + \sum_p[(p+1)/(p^2(p-1)^2) - 1/(p^2(p-1)^2)] = \mathcal{C}^2 + \sum_p p/(p^2(p-1)^2)$. Hence the **variance** equals

$$\mathcal{C}_2 - \mathcal{C}^2 = \sum_p \frac{1}{p(p-1)^2}.$$

This is a clean expression. Numerically, $\sum_p 1/(p(p-1)^2) = 1/2 + 1/12 + 1/80 + \cdots \approx 0.60132\ldots$

6. Extremal Behavior of the Error Term

From Theorem 3.1, the error $E(N) = S(N) - \mathcal{C}N$ is always negative and satisfies $E(N) = -\sum_{p\le N} s_p(N)/(p(p-1)) + O(1/\log N)$. We now analyze when $|E(N)|$ is extremal.

Proposition 6.1 (Maximizers of $|E(N)|$)**.** *Among $N \le X$, the error $|E(N)|$ is maximized (up to lower-order terms) when N has maximal digit sums in small primes. Specifically:*

(a) *For* $N = 2^k - 1$ *(binary repunit),* $|E(N)| \geq k/2 = (\log(N+1))/(2\log 2) \approx 0.721 \log N$.
(b) *More generally, for* $N = lcm(2^{a_2} - 1, 3^{a_3} - 1, 5^{a_5} - 1, \ldots)$ *(a number with maximal digit sums in many small bases simultaneously), one achieves* $|E(N)| \geq (\kappa/2 - \varepsilon) \log N$ *for any* $\varepsilon > 0$.

Conversely, $|E(N)|$ *is minimized when* N *is a prime power* p^k*, since* $s_q(p^k)$ *is typically small for* $q \neq p$ *and* $s_p(p^k) = 1$.

Proof. Part (a): For $N = 2^k - 1$, we have $s_2(N) = k$, so the contribution from $p = 2$ alone is $k/(2 \cdot 1) = k/2$. All other prime contributions are non-negative, so $|E(N)| \geq k/2 + O(1/\log N)$.

Part (b): By Dirichlet's theorem on simultaneous approximation, we can find $N \leq X$ such that $s_p(N) \geq (1 - \varepsilon)(p - 1) \log_p N$ for all primes $p \leq Y$, giving $|E(N)| \geq (1-\varepsilon) \sum_{p \leq Y} \log N/(2p \log p) + O(1)$. Letting $Y \to \infty$ slowly (e.g., $Y = \log\log X$) yields the claim.

For the minimizer claim: if $N = p^k$, then $s_p(N) = 1$, and for $q \neq p$, $s_q(p^k) = O(k)$ but typically of order $k(q-1)/(2\log_q p)$, which is moderate. The dominant saving comes from the base-p digit sum being 1. □

7. A Convolution Identity

The Dirichlet series representation (11) implies arithmetic identities via Dirichlet convolution.

Theorem 7.1. *The logarithmic arithmetic derivative can be expressed as*

$$\ell(n) = \frac{D(n)}{n} = \sum_{d|n} \Lambda_\star(d), \tag{15}$$

where $\Lambda_\star$ *is the arithmetic function defined by*

$$\Lambda_\star(n) = \begin{cases} 1/p & \text{if } n = p^k \text{ for some prime } p \text{ and } k \geq 1, \\ 0 & \text{otherwise.} \end{cases} \tag{16}$$

Equivalently, $\ell = \mathbf{1} * \Lambda_\star$ *in the Dirichlet convolution ring, where* $\mathbf{1}(n) = 1$ *for all* n.

Proof. For $n = p_1^{a_1} \cdots p_r^{a_r}$,

$$\sum_{d|n} \Lambda_\star(d) = \sum_{i=1}^{r} \sum_{k=1}^{a_i} \frac{1}{p_i} = \sum_{i=1}^{r} \frac{a_i}{p_i} = \ell(n).$$ □

Corollary 7.2. *By Möbius inversion,* $\Lambda_\star(n) = \sum_{d|n} \mu(n/d)\, \ell(d)$.

Remark 7.3. The function $\Lambda_\star$ is a "weighted" von Mangoldt function: $\Lambda_\star(p^k) = 1/p$ compared to $\Lambda(p^k) = \log p$. The convolution identity $\ell = \mathbf{1} * \Lambda_\star$ is the arithmetic-derivative analogue of $\log = \mathbf{1} * \Lambda$ from classical analytic number theory. This parallel is reflected in the Dirichlet series: $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)/n^s$ corresponds to $F(s) = \zeta(s) \cdot \sum \Lambda_\star(n)/n^s$.

8. The Dirichlet Series of $D(n)$ Itself

Theorem 8.1. *For $\Re(s) > 2$,*

$$\sum_{n=1}^{\infty} \frac{D(n)}{n^s} = \zeta(s-1) \sum_{j=1}^{\infty} P(j(s-1)+1). \tag{17}$$

This has a simple pole at $s = 2$ with residue $\mathcal{C}$, confirming

$$\sum_{n=1}^{N} D(n) \sim \frac{\mathcal{C}}{2} N^2.$$

Proof. Since $D(n) = n\,\ell(n)$, we have $\sum D(n)/n^s = \sum \ell(n)/n^{s-1} = F(s-1)$ for $\Re(s-1) > 1$, i.e., $\Re(s) > 2$. The formula follows from Theorem 4.1. The pole at $s = 2$ corresponds to the pole of $\zeta(s-1)$ at $s = 2$, and the Wiener–Ikehara theorem gives $\sum_{n \le N} D(n) \sim \mathcal{C}\, N^2/2$. □

9. Computational Verification

We implemented the arithmetic derivative and verified all theorems computationally. Table 1 shows selected values of $S(N)$, $\mathcal{C}N$, and the error $E(N)$.

Table 1. Computational verification of the asymptotic $S(N) \approx \mathcal{C}N$ and the one-sided bound $E(N) < 0$.

N	$S(N)$	$\mathcal{C}N$	$E(N)$	$E(N)/\log N$
10	5.876	7.732	−1.856	−0.806
10^2	74.207	77.316	−3.109	−0.675
10^3	767.76	773.16	−5.393	−0.781
10^4	7725.8	7731.6	−5.756	−0.625
10^5	77,308	77,316	−7.35	−0.639

All values of $E(N)$ are strictly negative, confirming Theorem 3.1. The ratio $E(N)/\log N$ remains approximately -0.7, consistent with $E(N) = \Theta(\log N)$ from Theorem 3.2.

We also verified the Legendre identity (Theorem 2.1) exactly for all $N \le 10^4$ (to machine precision), and the Dirichlet series identity (Theorem 4.1) by comparing truncated Euler products against direct summation for $s = 2, 3, 4, 5$ to at least 10 decimal places.

The variance constant $\mathcal{C}_2 - \mathcal{C}^2 = \sum_p 1/(p(p-1)^2)$ was computed to be $0.60132\ldots$ by summing over the first 10^6 primes. The empirical variance $\frac{1}{N}\sum_{n \le N}(\ell(n) - \mathcal{C})^2$ at $N = 10^5$ was $0.6010\ldots$, in excellent agreement.

10. Concluding Remarks

We have established a suite of exact and asymptotic results for the logarithmic arithmetic derivative via a simple but apparently overlooked connection to Legendre's formula for $v_p(N!)$.

Several questions remain open:

(1) **Finer asymptotics.** Can one obtain a full asymptotic expansion of $S(N)$ beyond the $\mathcal{C}N + O(\log N)$ result? The digit-sum error $\sum_p s_p(N)/(p(p-1))$ is not a smooth function of N, and its distributional properties deserve further study.
(2) **Distribution of $\ell(n)$.** Is $\ell(n)$ normally distributed over $\{1, \ldots, N\}$ in the sense of Erdős–Kac? Since $\ell(n)$ is a sum of "independent" terms $v_p(n)/p$ over primes, one expects a central limit theorem, but the weights $1/p$ decay, and the effective number of contributing primes is $O(\log\log N)$, requiring careful analysis.
(3) **Higher-order correlations.** What is the behavior of $\sum_{n \le N} \ell(n)\,\ell(n+h)$ for fixed h? This would connect the arithmetic derivative to the additive structure of the integers.
(4) **Connections to L-functions.** The Dirichlet series $F(s) = \zeta(s)\,G(s)$, where $G(s) = \sum_{j \ge 1} P(js+1)$ involves the prime zeta function, suggests potential connections to the distribution of primes and possibly to zeros of $\zeta(s)$ through the analytic properties of G.

References

[1] E. J. Barbeau, Remarks on an arithmetic derivative, *Canad. Math. Bull.* **4** (1961), 117–122.
[2] V. Ufnarovski and B. Åhlander, How to differentiate a number, *J. Integer Seq.* **6** (2003), Article 03.3.4.
[3] P. Haukkanen, J. K. Merikoski, and T. Tossavainen, On arithmetic partial differential equations, *J. Integer Seq.* **19** (2016), Article 16.8.6.